UNIT INVENTORY DATA BASE DEFINITIONS

"Unit Name" (No field number): Listed under "Unit Name" is the identification of the ground unit for which data is recorded in the Unit Inventory Data Base record. Data is recorded for three echelons: division (or independent armored brigade-) level; corps-level, and army-level.

Unlike records in the Unit Data Base (see the ACSDB Unit Data Base narrative), records for US and German division- or brigade-level units in the Unit Inventory Data Base are not subdivided into two parts with different identifications, i.e., "28th ID" and "28th ID Att." Instead, all Unit Inventory records pertaining to divisions, including those describing the equipment of the non-divisional units attached to divisions, are identified only by the division (or brigade) name, and the equipment of the division (or brigade) itself and any attachments to it are recorded on different pages. For a comprehensive explanation of this identification system, see the narratives on the derivation of the US, German, and British inventory data.

"Date" (No field number): Provided under "Date" is the date in month-day-year format of the record. See the ACSDB Time Conventions narrative for definitions of time periods for the US, British, and German unit records in the Unit Inventory Data Base.

"Nationality" (No field number): This is the nationality of the unit for which data is recorded. "A" designates American (US), "B" designates British, and "G" designates German.

"Amt" (Fields 1-16): Recorded under "Amt" (amount) are the types and numbers of items of equipment on-hand and in operational status. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Dst" (Fields 1-16): Recorded under "Dst" (destroyed) are the types and numbers of items of equipment destroyed in action. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Dmg" (Fields 1-16): Recorded under "Dmg" (damaged) are the types and numbers of items of equipment damaged in action. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Abd" (Fields 1-16): Recorded under "Abd" (abandoned) are the types and numbers of items of equipment abandoned in action. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Rpl" (Fields 1-16): Recorded under "Rpl" (replacement) are the types and numbers of items of equipment received as replacement by a unit. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Rpr" (Fields 1-16): Recorded under "Rpr" (in-repair) are the types and numbers of items of equipment in short- or long-term repair, or undergoing maintenance. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Rtn" (Fields 1-16): Recorded under "Rtn" (returned) are the types and numbers of items of equipment returned to operational status from repair. This field is used primarily to describe the status of armored fighting vehicles and artillery weapons, as discussed in the narratives on US, British, and German inventory data derivation. Note that equipment designations match those used in the ACSDB T/O&E and Weapons Data Bases.

"Remarks" (No field number): Entered under "Remarks" is information in text format which qualifies and amplifies data in any other field in the record.

"Sources" (No field number): Due to the considerable amount of estimation and the variety and number of sources required to generate data for the Unit Inventory Data Base, this paper serves as the source reference for the Unit Inventory Data Base. To record in Reference Data Base format all sources and data estimation used for the Unit Inventory Data base would have proved impractical and unwieldy. The narratives on the derivation of data for US, British, and German inventory data discuss in detail the sources and estimation processes employed to generate that data.

OPERATIONAL AND REPAIR STATUS FOR ARMORED FIGHTING VEHICLES IN THE ACSDB

In examining US National Archives records on US tank and tank destroyer units, it was found that the following definitions for vehicle status were most often used:

- Operational: Operational or available for operations in less than 24 hours.
- Repair: Non-operational with an estimated repair time of greater that 24 hours.

"Repair" status evidently included all types of repair due to battle damage or routine maintenance. All vehicles recovered from the battlefield with battle damage were generally classified as "repair" until and unless the vehicle was declared a battle loss by Ordnance. Thus, a vehicle recovered by unit ordnance recovery teams could be evacuated to 3d or 4th echelon repair and still be carried as a "repair" on the unit's vehicle status report. For the purposes of the ACSDB, all vehicles carried on the reports as "operational" are considered on-hand ("Amt"), all listed in the reports as "repair" are considered in-repair ("Rpr"), and, in the absence of other information, a drop in "repair" without an increase in "operational" in the reports is considered a destroyed vehicle ("Dst").

No precise definitions of vehicle status for the German Army were found in research. However, the most common conventions followed appear to be:

- Operational (ready for action).
- Short-term repair (ready for action in less than three weeks).
- Long-term repair (ready for action in more than three weeks).

As with the US terminology, "repair" included both routine maintenance and battle damaged vehicles. Virtually all German maintenance and repair in the Ardennes Campaign appears to have been conducted by division-level maintenance units and below. The six tank repair workshops available to German Army Group B apparently functioned as a parts depot and distribution network, due to a lack of sufficient numbers of tank transporter vehicles and motor fuel. The workshops cannibalized parts from new (replacement) vehicles and forwarded them to the divisional workshops for use in repairing vehicles in divisional repair.

US National Archives Foreign Military Study ETHINT 61, An Interview with Gen Pz Horst STUMPFF: TANK MAINTENANCE IN THE ARDENNES OFFENSIVE, discusses in general terms the German tank maintenance organization during the Ardennes Campaign.

US Equipment Inventory Data

INTRODUCTION

This narrative serves as the bibliographic reference for sources used in generating equipment inventory data for US units. It also explains the data estimation methodologies used to generate this data, as well as any peculiarities inherent to the data, such as the system of page numbers used in recording data for the numerous US Army divisional and non-divisional units which participated in the Ardennes Campaign. The narrative should be reviewed thoroughly and in conjunction with the Ardennes Campaign Simulation Data Base (ACSDB) Unit Inventory Data Base narrative in order to understand most fully the US data recorded in the Unit Inventory Data Base.

ORGANIZATION OF THE US INVENTORY DATA BASE RECORDS

There are over 9,000 records in the ACSDB Inventory Data Base used to record equipment inventory data on US units which participated in the Ardennes Campaign. The records contain data on equipment with infantry, airborne, and armored divisions, non-divisional combat units, and non-divisional service support units. The key to understanding in which records the data of the various units is recorded is the page number of the record as described in the following scheme.

Unit Name	Page #	Description
ID	1	All equipment of the three organic infantry regiments and one organic cavalry reconnaissance troop of an infantry division.
AD (Heavy)	1	Miscellaneous equipment of the two organic armored regiments of a heavy armored division, including M-3 HT, M-21 81mm, M-2 .50 HMG, 1 ton trl, 10 ton wrecker, .25 ton 4x4, and 2.5 ton 6x6. Also, all tanks in the division (the various M-4 models and the M-5 Tk), all M-8 AC, all M-32, and all M-8 75mm SP.
AD (Standard)	1	Miscellaneous equipment of the three organic tank battalions of a standard armored division, including M-9 2.36" Rkt, M-21 81mm, M-3 HT, .25 ton 4x4, 2.5 ton 6x6, 10 ton wrecker, and 1 ton trl. Also, all tanks in the division (the various M-4 models and the M-5 Tk), all M-8 AC, all M-32, and all M-8 75mm SP.

AbnD	1	All equipment of the organic parachute and glider infantry regiments of an airborne division. With the 82d and 101st Airborne Divisions, this equipment includes that of one glider and two parachute infantry regiments. With the 17th Airborne Division, this equipment includes that of one parachute and two glider infantry regiments.
ID	2 & 3	All equipment of all organic infantry division units not recorded on Page 1.
AD (Heavy)	2	All equipment of the one organic armored infantry regiment and one organic armored reconnaissance battalion of a heavy armored division, except the reconnaissance battalion's M-8 AC, M-8 75mm SP, and M-5 Tk which are included on Page 1.
AD (Standard)	2	All equipment of the three organic armored infantry battalions and one organic cavalry reconnaissance squadron of a standard armored division, except the reconnaissance squadron's M-8 AC, M-8 75mm SP, and M-5 Tk which are included on Page 1.
AbnD	2	All equipment of all organic airborne division units not recorded on Page 1.

AD (Heavy and Standard)	3 & 4	All equipment of all organic armored division units not recorded on Pages 1 & 2.
ID, AD, AbnD, Corps Troops, and Army Troops	5	All artillery guns and howitzers, antiaircraft artillery weapons (including light pieces), and 4.2-inch chemical mortars of non-divisional field artillery, antiaircraft artillery, and chemical mortar units attached to divisions or subordinated to corps or armies.
ID, AD, AbnD, Corps Troops, and Army Troops	6	Selected equipment of non-divisional tank destroyer, tank, and cavalry units attached to divisions or subordinated to corps or armies. This equipment includes -- for tank destroyer units -- M-10, M-18, and M-36 tank destroyers, and M-5 3" AT (M-5 antitank) guns. For cavalry units, it includes M-5 Tk, M-8 AC, and M-8 75mm SP. For tank units, it includes the various M-4 models, M-5 Tk, M-24, and M-32. Equipment in records numbered as Page 6 are recorded in this order: tank destroyer unit equipment, cavalry unit equipment, and tank unit equipment. Therefore, there is some duplication of equipment names such as M-5 Tk, which is found in both cavalry and tank units, in cases where both kinds of units are subordinated to a division, corps, or army.

ID, AD, AbnD, Corps Troops, and Army Troops	8 & 9	Miscellaneous equipment not recorded on Pages 5 & 6, belonging to all non-divisional field artillery, antiaircraft artillery, chemical mortar, tank destroyer, tank, cavalry, airborne infantry, armored infantry, combat engineer, ranger, and other combat units.
Corps HQ and Army HQ	8 & 9	All equipment of non-divisional service support units subordinated to corps or army.

Note: ID = Infantry Division, AD = Armored Division, and AbnD = Airborne Division. The 2d and 3d ADs were "heavy" armored divisions. The 4th, 5th, 6th, 7th, 9th, 10th, and 11th ADs were "standard" armored divisions.

The "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base records the daily order of battle of US combat units during the Ardennes Campaign. The order of battle serves as the "blueprint" for determining equipment strengths in records numbered from Pages 5 to 9. For example, if the 28th ID has one non-divisional tank battalion attached to it on 25 December 1944, the equipment strengths of the tank battalion, as well as any attrition sustained by the tanks of the battalion, are reflected in the equipment recorded on Pages 6, 8, and 9 for the unit identified as "28th ID" on 12/25/44.

Similarly, for Corps HQ and Army HQ units, the equipment recorded on Pages 8 & 9 for units identified with these designations reflects the equipment strengths of non-divisional service support units. The daily order of battle of these units is not recorded in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base, but instead is described in the US Non-Divisional Personnel Data narrative.

The system used for recording US inventory data, as described above, was created to facilitate generation, computer entry, and proofing of the data, and to allow for the data's logical, systematic presentation in computer format. The complications encountered in generation of data for units whose subordination frequently changes on a daily basis are described

in the ACSDB Unit Data Base narrative. These complications are compounded by the necessity of recording data for over 200 different items of equipment. The system employed to record US inventory data was created after considerable deliberation. It allows for an organized and logical presentation of the data. Short of recording data for all separate non-divisional units, a task of unmanageable data entry and storage proportions, it most faithfully represents the daily status of equipment for US Army units in the Ardennes Campaign.

US INVENTORY DATA GENERATION

Unlike the US Army unit personnel data available in primary source records, especially the daily personnel statistics found in the US 12th Army Group G-1 Daily Summary (located in US National Archives Record Group 407, Boxes 1753 and 1754), US Army primary source records lack the detailed, daily information on equipment status necessary to generate data at the ACSDB's required level of detail. Therefore, it proved necessary to employ several estimation methodologies to fill in missing data points and to supplement the data available in primary source records.

The principal estimation methodology uses an equipment attrition rate based on personnel attrition, which is applied to the equipment of the organic divisional "line" units found on Page 1 for infantry and airborne divisions, and Page 2 for armored divisions. This methodology is discussed below in greater detail.

Another frequently-used estimation methodology may be more accurately described as an application of mathematics, based on an underlying assumption. The underlying assumption is that the combat support and service support units of US Army divisions, corps and armies, including field and antiaircraft artillery units but not armored units, incurred minimal equipment losses and that their equipment strength was generally at or near full authorization levels. This assumption is based on the limited information found in research on non-divisional units. (To reduce the scope of a considerable data assembly task, research in US Army records for the ACSDB was directed at units at the division-level and above, except in the case of separate non-divisional armored units, as discussed below.) Based on the above-described assumption, calculation of inventory data for all units recorded on Pages 6, 8, and 9; for infantry and airborne divisions on Pages 2 and 3; and for armored divisions on Pages 3 & 4 essentially involves multiplying the numbers of particular types of unit by the authorized equipment strengths of the units, with the numbers of particular unit types determined by the daily order of battle. Exceptions to this approach do occur. For example, in the cases of the non-divisional field artillery battalions which lost unusually high amounts of equipment in the initial German attacks of the Ardennes Campaign, the actual equipment losses of these units are recorded, if known, or otherwise estimated, and their equipment strengths decreased accordingly. Also, in cases in which historical data is available, it is used in lieu of full authorized equipment strengths. These cases are described below in greater detail.

For all armored units (armored divisions, non-divisional tank units, towed and self-propelled tank destroyer units, and non-divisional cavalry units), most primary source documentation was researched in order to assemble as much information as possible on the equipment status of these units. These include the records of the armored divisions, the non-divisional battalions, and tank status reports. The sources researched and their use in generating daily armored fighting vehicles strengths as recorded on Page 6 for all units and Page 1 for armored divisions are discussed below in greater detail.

British Equipment Inventory Data

British equipment inventory data for the Ardennes Campaign Simulation Data Base (ACSDB) was derived from the few available primary source records with information on British equipment data and the authorized equipment strengths of units. This paper serves as the bibliographic reference for sources used to generate British equipment inventory data and also describes the general methodologies used to estimate data that was not available in sources. All sources mentioned in the paper are fully referenced in Attachment 1 of the paper.

British Armored Fighting Vehicles.

The only complete armored fighting vehicle (AFV) status report extant for British units during the Ardennes Campaign period is for 28 December 1944, as found in WO171/346. This gives a breakdown of major AFV types ready for action, in repair, and in reserve. The only other report available is the daily AFV states given by the 29th Armoured Brigade in WO171/627 and WO171/4345. This gives an aggregate ready for action figure for main battle tanks only on a daily basis. Occasional mention is also given in various unit diaries of tank losses, as in the case of the 29th Armoured Brigade. In the case of the 29th Armoured Brigade, these seem to follow the fluctuations found in the daily reports. Daily in-repair states by AFV type -- aggregated at corps -- are also found in WO171/348 and WO171/4084. From the daily unit diary entries it appears that the Guards Armoured Division and 33d Armoured Brigade suffered no vehicle losses; it was thus assumed that the AFV strengths in the report for 28 December would be in effect for the entire Ardennes Campaign period in question. For the 34th Tank Brigade the corps repair states were used to generate the daily strength figures with 28 December as a benchmark. This proved possible as the 34th Tank Brigade was the only formation in the Ardennes Campaign equipped with the various models of Churchill tank. The 29th Armoured Brigade figures were generated from the 28 December report with losses as given in the daily unit diary entered for the relevant date.

British Artillery.

WO171/231 and WO171/3964 give daily tube strengths for each corps in the Second British Army on a daily basis. These match the authorized T/O&E strengths of the corps artillery in all cases. It is not known if this report is for all tubes available or for all tubes in service. In the absence of evidence to the contrary, all British artillery is assumed to be at full strength throughout the period.

Other Items of Equipment.

As no data was found in research for equipment types other than AFVs and artillery, it was assumed that British units started on their first day of participation in the Ardennes Campaign at full T/O&E strength. The other equipment types were then attritted by matching similar British systems with those US systems found in the US units in the Unit Inventory Data Base and applying the same attrition methodology as used for the US, i.e., the "Normal" equipment loss rate described in the "US LINE INFANTRY BATTALION AND REGIMENT DATA" section of the US Equipment Inventory Data narrative.

Format Used for Recording British Equipment Inventory Data.

Equipment data for British units is recorded using a numbering sequence similar to those employed for US and German units, based on page numbers and unit designations. The format used for the British equipment inventory as follows:

Unit Designation	Page(s)	Description of Data
29th ArmBde	1-2	Data for organic units of 29th ArmBde
29th ArmBde Att	1-2	Data for units attached to 29th ArmBde
33d ArmBde	1-2	Data for organic units of 33d ArmBde
33d ArmBde Att	1	Data for units attached to 33d ArmBde
34th TkBde	1	Data for organic units of 34th TkBde
34th TkBde	1	Data for units attached to 34th TkBde
43d ID	1-2	Data for organic units of 43d ID
43d ID Att	1	Data for units attached to 43d ID
51st ID	1-2	Data for organic units of 51st ID
51st ID Att	1	Data for units attached to 51st ID
53d ID	1-2	Data for organic units of 53d ID
53d ID Att	1	Data for units attached to 53d ID
Gds AD	1-3	Data for organic units of Guards AD
Gds AD Att	1	Data for units attached to Guards AD
6th AbnD	1-2	Data for organic units of 6th AbnD
6th AbnD Att	1	Data for units attached to 6th AbnD

XXX Corps HQ	1	Data for non-divisional service support units subordinated directly under the XXX Corps
XXX Corps Troops	1-2	Data for non-divisional combat units subordinated directly under the XXX Corps

Note: AbnD = airborne division; AD = armored division; ArmBde = armoured brigade; ID = infantry division; TkBde = tank brigade; Att = attachments; and HQ = headquarters.

Attachment 1

Sources Used for British Equipment Inventory Data

WO171/231 -- Second Army "Q" (Quartermaster) War Diary, Dec 44.

WO171/346 -- XXX Corps War Diary, 15-31 Dec 44.

WO171/348 -- XXX Corps "Q" (Quartermaster) War Diary, Dec 44.

WO171/378 -- Guards Armoured Division War Diary, 18-30 Dec 44.

WO171/627 -- 29th Armoured Brigade War Diary, 1-31 Dec 44.

WO171/3964 -- Second Army "Q" (Quartermaster) War Diary, Jan 45.

WO171/4084 -- XXX Corps "Q" (Quartermaster) War Diary, Jan 45.

WO171/4345 -- 29th Armoured Brigade War Diary, 1-31 Jan 45.

German Equipment Inventory Data Derivation

INTRODUCTION

German equipment inventory (weapons and vehicles) strengths and losses for the Ardennes Campaign Simulation Data Base (ACSDB) were estimated in a multi-step process using all available primary and secondary sources. As with all elements of the German data relevant to the ACSDB, the German equipment inventory lacked documentation with the comprehensive, daily statistics necessary to meet the detailed requirements of the Unit Inventory Data Base. Records for only a relatively small number of units (several LXXXI Korps divisions and the 116th PzD in December) contained information on actual daily losses of equipment. Therefore, it was necessary to interpolate and extrapolate most of the required data from the periodic reports (generally monthly) of the divisions for which this data was available, and then use the experiences of these divisions to estimate equipment losses for those divisions lacking any documentation.

The sources used for German equipment inventory estimation consisted primarily of monthly divisional reports, several corps reports, miscellaneous Army Group B artillery and tank reports, and periodic Inspector General of the Panzer Troops reports. These latter documents (15 and 30 December and 15 and 30 January) contained information on operational and in-repair armored fighting vehicle strengths by unit (including self-propelled antiaircraft, antitank, and artillery systems).

The Army Group B reports proved most useful for German 16 December artillery and armored fighting vehicle strengths. Monthly divisional reports were available almost exclusively for army and SS panzer (armored) and panzer grenadier (armored infantry) divisions. These reports were in general either divisional Zustandsbericht (situation reports) or Gliederung (organizational reports). The former document contained information on operational and in-repair major items of equipment, but usually grouped together all equipment types, such as artillery, under one general category without differentiation between specific weapons. The Gliederung was a report in graphic format roughly equivalent to a table of equipment, which gave the actual strength of the divisions, at a level of detail usually by specific weapons, but without any differentiation between operational and in-repair status.

Most sources referenced in this paper were obtained from the US National Archives or the German Archives in Freiburg, West Germany. For convenience and to facilitate preparation of this

paper, sources are referenced in abbreviated format (a source with a "T" prefix was obtained from the US National Archives, a source with an "RH" prefix from the German Archives). Full bibliographic annotations of all sources referenced in this paper are provided in the Bibliography Data Base.

EQUIPMENT INVENTORY DATA GENERATION METHODOLOGY

Using the various sources available for certain divisions, it was possible to determine their estimated periodic equipment strengths. The sources used for these divisions and their application in the estimation process are described and referenced in one- or two-page individual descriptions for all divisions in a later section of this paper.

The overall approach used to estimate German equipment inventory was as follows:

1) Estimate armored fighting vehicle (including self-propelled antitank and assault guns) inventory data for all German divisions and non-divisional units in the ACSDB. This task is explained in the German Armored Fighting Vehicle Data Generation Methodology narrative.

2) Determine the equipment inventory start strengths (all other equipment besides tanks) for all German divisions in the ACSDB. The sources used and their application are explained in the individual division descriptions.

3) Estimate daily equipment strengths and attrition (all other equipment besides armored fighting vehicles) of divisions for which sufficient periodic reports (Gliederung, Zustandsbericht, etc.) existed. The six divisions addressed in this step were German Army panzer and panzer grenadier divisions: PzLehrD, 2d PzD, 9th PzD, 116th PzD, 3d PzGD, and 15th PzGD. This process was keyed to personnel attrition and is explained in greater detail below as the "Army Panzer Divisions averages."

4) Using averages derived from the experiences of the six panzer and panzer grenadier divisions, estimate daily equipment strengths and attrition of divisions for which no or minimal data was available. The divisions addressed in this step were the SS panzer divisions and the German infantry-type divisions (fallschirmjaeger (paratrooper), volks grenadier, and infantry), which lacked periodic documentation of equipment strength. This process is described in greater detail below.

5) Using the same averages employed in step 4), estimate equipment data of German non-divisional units. This process is also described below.

Active/Inactive/Out of Contact Assessments.

To estimate daily equipment strengths and losses of the six divisions for which the most extensive equipment data was available, a methodology was employed involving the activity levels and divisional battle casualties estimated in the German personnel data estimation process.

An example of its application is as follows: The PzLehrD's 1 December Zustandsbericht figure for light and heavy machine guns (LMG/HMG) strength was 538, and its 1 January figure 401. (Both sources are from German Archives record #RH10/172, pgs. 47 and 51.) These figures were used as the PzLehrD's 16 December and 1 January LMG/HMG strengths. The difference between them (137) was estimated as the net LMG/HMG losses for 16-31 December. To determine the daily losses, the battle casualties for 16-20 December (154) and 21-31 December (1,013) were used to determine percentages to apply to the net LMG/HMG losses to estimate losses for the two time periods--13 percent for 16-20 December (154/1,167), and 87 percent for 21-31 December (1,013/1,167). Thus, 13 percent of the 16-31 December LMG/HMG losses, or 18 weapons, were estimated to have been lost from 16-20 December, and 87 percent, or 119 weapons, from 21-31 December.

To determine daily losses, the Active/Inactive/Out of Contact assessments made for estimating personnel battle casualties were applied to the 16-20 and 21-31 December LMG/HMG losses. The basic underlying assumption of this approach was that equipment losses, like personnel losses, occurred at an estimated rate 6.5 times higher on "Active" Days than on "Inactive" Days, and that no losses occurred on the so-called "Out of Contact" Days. For the period 16-20 December the equation used to estimate losses on "Inactive" Days was: $a = 18/[(6.5 \times 3)+2]$, when a = losses on an "Inactive" Day, 18 = total 16-20 December losses, 3 = number of "Active" Days, and 2 = number of "Inactive" Days, or 0.84. Rounded off to the nearest whole number, this value equaled 1, and multiplied by 6.5, it equaled 5.46 (0.84 x 6.5), which (rounded off) was the estimated number of LMG/HMG losses on "Active" Days. Thus, the number of LMG/HMG losses on 16 and 18 December was estimated to be 1, and on 17, 19, and 20 December, 5, 5, and 6 respectively. Losses on "Inactive" Days from 21-31 December were estimated to be 2-3/day -- $119/[(6.5 \times 7)+4]$, and 15-16/day on "Active" Days (6.5 x 2.4). Note that this process was applied for four time periods of the Ardennes Campaign, i.e., 16-20 December, 21-31 December, 1-10 January, and 11-16 January. This was done because the manner in which German battle casualites were estimated for the ACSDB used "Active/Inactive/Out of Contact" assessments for each of the four time periods. Therefore, in order to derive evenly weighted equipment losses based on personnel attrition, the equipment losses were also estimated for each of the four periods.

The example presented here represents a straightforward application of the methodology. It involves a weapon for which only on-hand strength data was required (see details below on these requirements). For weapons which required determination of daily destroyed, damaged, abandoned, and in-repair status data, comparison was made of the periodic reports which differentiated between operational and in-repair equipment (Zustandsbericht, for example) or judgement was frequently used to estimate the data required for other than on-hand status. A detailed pictorial account of the Ardennes Campaign, Jean Paul Pallud, Battle of the Bulge: Then and Now (London: After the Battle, 1984), proved particularly useful in determining some daily data on destroyed and abandoned weapons and vehicles, as it did for German tanks. When the total number of equipment losses of a particular weapon was substantially less than the number of days for the period in which the losses occurred (an estimate of only one loss over a 10-day period, for example), the day of the loss was determined, when possible, using secondary sources with information on a units' daily activities, or, when no information was available, selected at random.

The estimation methodology outlined above was used in conjunction with other information (such as Pallud's book and any daily loss reports such as those of the 116th PzD for several days in December 1944) to determine all equipment data for the PzLehrD, 2d PzD, 9th PzD, 116th PzD, 3d PzGD, and 15th PzGD. This data was then combined to determine averages of destroyed and damaged equipment and return from repair and replacement equipment for use in estimating SS panzer and infantry-type divisions' equipment losses. These averages, termed the "Army Panzer Divisions averages," are presented in Attachments 1-4.

The Army Panzer Divisions averages.

Equipment attrition of German divisions other than the six Army armored units described above, necessitated the use of averages derived from the experience of the six Army units. This was due to the lack of periodic equipment status reports for the non-Army armored divisions, particularly the infantry-type divisions, but also for the SS panzer (armored) units.

An example of the application of the averages is as follows: The 212th VGD was estimated to have sustained 1,662 battle casualties between 16 December and 16 January. The average rate at which the six Army panzer divisions lost light and heavy machine guns (LMG/HMG) was estimated to be 9.597 per 100 battle casualties. Therefore, the 212th VGD's total LMG/HMG losses were estimated to be 9.597 x 16.62, or 160 weapons. The numbers lost on "Active" and "Inactive" Days were estimated using the Active/Inactive/Out of Contact assessments. On an "Inactive" Day, the number was estimated as follows: $a = 160/(6.5 \times 6)+26$, where a = the number of LMG/HMG losses on an "Inactive" Day, 6 =

the number of "Active" Days, and 26 = the number of "Inactive" Days, or 2-3/day (2.46/day). The number lost on an "Active" Day was estimated to be 6.5 times greater than 2.46, or 15.99 (16).

For the same division, 75mm antitank gun (PAK75) losses and return from repair weapons were estimated as follows: 16.62 x 0.390, or 6 destroyed, 16.62 x 0.042, or 1 damaged, and 26 x 0.104, or 3 returned from repair (26 equaling the number of "Inactive" Days). Note that equipment loss estimates (destroyed and damaged) were keyed to personnel battle casualties, whereas equipment gains were keyed to numbers of "Inactive" Days. Since the total number of damaged PAK75s did not exceed one weapon, the number of returned from repair was limited to one. As no secondary source information was available on PAK75 losses, the day on which the one PAK75 was returned from repair was selected at random. (This is a typical example of random assignment of equipment strength changes. A total of 1 PAK75 was estimated to have been returned from repair over a 32-day period, 26 of which were assessed as "Inactive" Days. With no other information available, the day randomly selected for the return from repair of the one damaged PAK75 was 5 January.)

The "Army Panzer Divisions averages" were used in conjunction with whatever other data or information on equipment was available in primary or secondary sources. For example, the Inspector General of the Panzer Troops reports provided information for many units' 30 December armored fighting vehicles' status (half-tracks, armored cars, etc.). This data would be used with 16 December strengths to estimate the 16-29 December losses, employing the Active/Inactive/Out of Contact assessments, while the 30 December-16 January losses would be estimated using the Army Panzer Divisions averages.

Details on equipment strength and attrition estimation for all 39 divisions and the three brigades involved in the Ardennes operation are provided in the individual division descriptions in a later section of this report. References of all sources consulted and the various methodologies employed are also provided.

The 340th VGD Vehicle averages.

Upon application of the Army Panzer Divisions averages to the German infantry-type divisions, it was determined that the average numbers lost per 100 battle casualties for certain non-armored vehicles were so high that the divisions' inventory of these vehicles was depleted. (Those vehicles included wheeled personnel carriers, cargo trucks, two models of prime movers, and horse-drawn vehicles---PKW, LKW, RSO, MAUL, and HDR.) Therefore, alternative averages were derived for these vehicles, from the experience of one volks grenadier division for which reliable secondary source data was available---the 340th VGD.

B678, pgs. 47-48 (source fully referenced in the Bibliography Data Base), provided information on the 340th VGD's 22 November-27 January equipment losses (42 PKW, 38 LKW, 22 RSO, 4 MAUL, and 346 HDR). Between 22 November and 27 January the division was estimated to have had 20 "Active" Days, 35 "Inactive" Days, and 12 "Out of Contact" Days. The corresponding number of days for 16 December-16 January were 8, 17, and 7. Using these numbers, total losses for 16 December-16 January were estimated. The total activity level "value" for 22 November-27 January was calculated as (6.5 x 20)+35, or 165, where 20 = the number of "Active" Days and 35 = the number of "Inactive" Days. For 16 December-16 January, this value was calculated to be 69, or (6.5 x 8)+17. The ratio of 69/165 equals 0.418, which value was applied to the total losses of each vehicle type to determine the estimated 16 December-16 January losses. The resulting values 17.556 (PKW), 15.884 (LKW), 9.196 (RSO), 1.672 (MAUL), and 144.628 (HDR), were used with the total 16 December-16 January divisional battle casualties (2,317) to calculate a loss rate per 100 battle casualties. For the vehicles listed above the values are as follows:

PKW:	17.556/23.17, or 0.758	loss/100	battle casualties
LKW:	15.884/23.17, or 0.686	"	"
RSO:	9.196/23.17, or 0.397	"	"
MAUL:	1.672/23.17, or 0.072	"	"
HDR:	144.628/23.17, or 6.242	"	"

These averages, termed the "340th VGD Vehicle averages," were used instead of the Army Panzer Divisions averages for volks grenadier, infantry, and fallschirmjaeger (VGD, ID, and FJD) non-armored vehicle losses.

Miscellaneous Notes on German Divisions' Equipment Data.

On-Hand Equipment. The fields in the Unit Inventory Data Base are broken down into seven major categories. These include: Amt (Amount On-Hand), Dst (Destroyed), Dmg (Damaged), Abd (Abandoned), Rpl (Replacement), Rpr (In-Repair), and Rtn (Return from Repair). Certain equipment was only tracked in the "Amt" field. This equipment included infantry weapons, antiaircraft weapons, trucks, prime movers, recovery vehicles, and horse-drawn vehicles. FLAK88 (the 88mm dual-purpose antiaircraft gun) were to be tracked in all categories, but it was determined that insufficient data was available in primary sources to estimate numbers of weapons in repair status or numbers of replacement weapons. Therefore, these weapons were tracked only in the "Amt" field.

Panzerfaust (PF) and Panzerschreck (PS). The panzerfaust (PF) was a German antitank weapon roughly equivalent in design to the modern day US Army M-72 LAW. Because it was a disposable weapon and because there is a general lack of data on its

"attrition," the start strength for this weapon for the divisions which fielded it is provided in the ACSDB, and that strength maintained throughout the 16 December-16 January period. Sufficient data on the German expenditure of this weapon in the Ardennes does not exist to estimate its daily on-hand strengths. The underlying assumption in maintaining the PF at its start strength is that the Germans were able to maintain a constant supply of this lightweight weapon with their units in the Ardennes.

The panzerschreck (PS), a German 88mm antitank weapon similar in design to the US Army "bazooka" (M-9 2.36" Rkt), was another weapon for which very little strength on loss data was available in German records. However, as it was not a disposable weapon, it was attrited in the ACSDB. The PS's attrition was estimated at the same rate as LMG/HMGs, on the assumption that the weapon was fielded with German front line combat units and therefore susceptible to loss and damage at rates similar to those of machine guns.

K75. The 75mm field gun (K75), a weapon used exclusively by the volks grenadier divisions (VGD) and volks artillery korps (VAK), was not a weapon in the panzer divisions' inventories. Therefore, no average was available in the "Army Panzer Divisions averages" for estimating its attrition. To address this shortfall, the Army Panzer Divisions averages of the 105mm howitzer (LFH105) were used for the K75 in the VGDs. Both the LFH105 and K75 were part of the VGD's artillery regiment equipment and served in a similar role as artillery, although the K75 had a secondary antitank capability.

German Non-Divisional Equipment Data Generation.

Few sources with daily data on German non-divisional combat units during the Ardennes Campaign were found in research. The general approach used to estimate data for non-divisional units was to use full authorized equipment strength of the unit as its start strength, and to attrit the equipment using the Army Panzer Divisions averages. To reflect shortages of transport vehicles with German units, the authorized strengths of the following systems were reduced to the indicated percentages:

- PKW	71%	of authorized strength
- LKW	84%	"
- MAUL	71%	"
- 1-5t PM	60%	"
- 8-18t PM	78%	"
- BGPZ-III/IV & V	67%*	"

* Based on BGPZ-V experience.

These reductions were made based on data found for the following units in the indicated sources. These were non-divisional units for which primary source data on equipment strengths at the beginning of December 1944 was available. The percentages of operational to authorized equipment strengths are show for items of equipment in the units' inventories.

Unit	PKW	LKW	MAUL	1-5t PM	8-18t PM	BGPZ-V
301st HyPzBN	90%	90%	83%	89%	70%	67%
319th PzCo	75%	100%	--	50%	--	--
341st StgBde	51%	82%	54%	--	67%	--
682d PjBNT	58%	51%	78%	--	90%	--
519th PjBN	79%	99%	69%	40%	86%	--
AVERAGE	71%	84%	71%	60%	78%	67%

Sources:

519th PjBN -- 1 Dec 44 Zustandsbericht (T314,1597,176)
682d PjBNT -- 9 Dec 44 Zustandsbericht (T314,1597,178)
341st StgBde -- 1 Dec 44 Zustandsbericht (T314,1597,164)
319th PzCo -- 4 Dec 44 Zustandsbericht (T314,1597,162)
301st HyPzBN -- 1 Dec 44 Zustandsbericht (T314,1597,159)

(All sources described in the Bibliography Data Base under "T314,1597,159-179.")

FORMAT FOR RECORDING GERMAN EQUIPMENT DATA IN THE UNIT INVENTORY DATA BASE

The variety of German units tracked in the Unit Inventory Data Base (divisions, non-divisional combat units, non-divisional service support units, etc.) necessitated use of a data recording format characterized by the employment of page numbers, a format similar to that used for US equipment data. (For details on the US format, see the US Equipment Inventory Data narrative.)

Equipment data of all organic divisional units of a division (i.e., the component units which comprise the division) was recorded on Pages 1-3 in the Unit Inventory Data Base. Note that the volks grenadier, infantry, and fallschirmjaeger (paratrooper) divisions generally required two pages per day (Pages 1-2) to record all equipment data of organic divisional units on, while army and SS panzer and panzer grenadier divisions required three pages (Pages 1-3), due to the greater variety of systems in the armored formations.

Included on the Pages 1-3 series, by convention on the last page of this series (Page 2 for volksgrenadier divisions, Page 3 for panzer divisions, etc.), was data on the armored fighting vehicles of any attached non-divisional panzer (armored) or self-propelled panzerjaeger (antitank) unit, most commonly independent heavy tank battalions (schwere panzer abteilung), antitank battalions (panzerjaeger abteilung), or assault gun brigades (sturmgeschuetz brigade). Only data on armored fighting vehicles of these units was included on the pages described above. Data on the other items of equipment of the non-divisional panzer and panzerjaeger units was recorded elsewhere in the Unit Inventory Data Base. The "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base records the daily order of battle of ground forces in the ACSDB. The order of battle identifies the panzer and panzerjaeger units and should be consulted to determine the parent units of the armored fighting vehicles on Pages 2 and 3.

Corps and army "troops" records ("5th PzArmy Troops," "15th Army Troops," "LXXIV K Troops," etc.) identified with Page 3 were also used to record the armored fighting vehicle data for non-divisional panzer and panzerjaeger units attached directly to corps or armies, as identified in the daily order of battle.

All other German equipment inventory data (i.e., equipment of all units other than divisions, except the armored fighting vehicles of non-divisional units) was recorded as follows:

- For equipment of divisional attachments (as identified in the order of battle), records identified as Page 7 and with the division's name ("9th PzD," "89th ID," etc.) were used.

Note that included here were the artillery and antiaircraft artillery weapons of any artillery or FLAK (antiaircraft artillery) units attached directly to divisions.

- For equipment of the non-divisional combat or service support units attached directly to corps or armies (as identified in the order of battle), records identified as Pages 1 and 2 and with the unit's name ("5th PzArmy," "I SSPzk Troops," "15th Army Troops," etc.) were used. (Due to a great variety of non-divisional units in some corps and armies, two records per day were sometimes required to record all equipment data.) Note that included here were the artillery and antiaircraft artillery weapons of any artillery or FLAK (antiaircraft artillery) units attached directly to corps or armies. Also note that equipment of immobile artillery and construction engineer units was recorded with the corps or army "HQ" unit ("6th PzArmy", "LXXXI K," etc.).

The ACSDB Unit Location Data Base

INTRODUCTION

The Unit Location Data Base of the Ardennes Campaign Simulation Data Base (ACSDB) is used to record daily information on the activities and locations of US, British, and German ground units at division-level and above. Four- or six-digit grid coordinates from 1943 maps produced by the Army Map Service are used to record the front-line trace and locations of other significant elements of units. Text fields in the Unit Location Data Base contain narrative material which describes the activities and missions of a unit, environmental conditions in the vicinity of a unit, status of unit commanders, etc.

This paper provides definitions of fields used in the Unit Location Data Base of the Ardennes Campaign Simulation Data Base (ACSDB). The paper also describes the general methodology used to derive location data for US, British, and German units. Because the location data was extracted from sources with relatively minimal estimation, unlike the personnel and equipment inventory data of the Unit and Unit Inventory Data Bases, it was possible to list all sources used for location data in records in the Reference Data Base. These Reference Data Base records are keyed to the Unit Location Data Base records as explained in a later section of this paper. Nevertheless, the major sources used for location data and their application in the ACSDB are described herein.

UNIT LOCATION DATA BASE DEFINITIONS

"Unit Name: (Field 1): Identification of the unit for which location data is recorded in the record is provided under "Unit Name." The designations used for divisions and brigades are the same ones used in the Unit and Unit Inventory Data Bases. For corps and armies, only the title of the unit is entered (i.e., "5th PzArmy," "XXX Corps," "1st Army," "III Corps," "I SSPzK," etc.), unlike the entries used in the Unit and Unit Inventory Data Bases which differentiate between corps/army headquarters and corps/army troops ("5th PzArmy Troops," "XXX Corps Troops," "1st Army HQ," etc.)

"Date" (Field 2): Provided under "Date" is the date in month-day-year format of the record. See the ACSDB Time Conventions narrative for definitions of time periods for the US, British, and German records in the Unit Location Data Base

"Nationality" (Field 3): This is the nationality of the unit for which data is recorded. "A" designates American (US), "B" designates British, and "G" designates German.

"Parent Unit" (Field 4): Recorded under "Parent Unit" is the identification of the immediate superior unit of the unit tracked in the Unit Location record. Designations recorded under "Parent Unit" are the same as those used for "Unit Name," i.e., "5th PzArmy," "I SSPzK," "XXX Corps," "1st Army," etc.

"HQ Starting Location" (Field 5): Recorded under "HQ Starting Location" is the location of the unit's headquarters at the beginning of the day. Its location is designated in all cases by the Universal Transverse Mercator (UTM) four- or six-digit grid coordinates used on the 1943 Geographical Section, General Staff maps produced by the Army Map Service. These maps are:

Sheet #	Name
8	Brussels
9	Liege
12	Namur
13	Marche
16	Mezieres
14	Arlon
R-1	Koeln
S-1	Bonn
T-1	Trier
V-1	Neunkirchen

If information is available, an actual place location name is included, most commonly a town, village, or city name.

"Mission" (Field 6): This is a text field, i.e., information recorded in this field is in text format. Under "Mission" is recorded a unit's basic mission for the day, including planned activities, operations, changes in organization, and other pertinent information. This is a "required" data field, i.e., information on unit mission is provided in the Unit Location Data Base for each unit for every day.

"Displacement" (Field 7): Under "Displacement" is recorded:

1) the straight-line distance in kilometers that the center of a unit's front line moved during the 24 hours beginning at 0600 hours of the date recorded under "Date," or

2) the straight-line distance in kilometers that the headquarters of a unit moved during the 24 hours beginning at 0600 hours of the date recorded under "Date," or

3) the straight-line distance in kilometers that the center of mass of a unit moved during the 24 hours beginning at 0600 hours of the date recorded under "Date," or

4) any other straight-line movement distance (in kilometers) moved by a unit from one point to another as specified under "Remarks."

The "Displacement" value is always positive. Displacement can indicate movement of a unit while in reserve or transit. Case 1) above is always used to describe unit movement while the unit is in a front-line position. The movement can be forward, lateral, or retrograde movement. Cases 2) to 4) above are most commonly used when a unit is in a reserve position, such as in transit to the front. The data recorded under "Displacement" in Cases 2) to 4) depends on the nature and availability of the information found in data sources. When not self-evident in meaning, data recorded under "Displacement" is always explained under "Remarks" to facilitate understanding of a unit's movement.

"Distance Opposed Advanced" (Field 8): Data recorded under "Distance Opposed Advanced" is the distance in kilometers that a unit moves in combat, and is only used to record data when a unit or significant elements of it are in the front line. "Distance Opposed Advanced" is a subset of location displacement, and its absolute value never exceeds the value of "Displacement." Positive values entered under "Distance Opposed Advanced" represent advances or forward movement, and negative values represent retrograde movement.

"Width of Front" (Field 9): Entered under "Width of Front" is a unit's front line width (in kilometers), bounded on its flanks by the grid coordinates provided under "Left Flank" and

"Right Flank" (Fields 10 and 11) and following a front line trace delineated by grid coordinates provided under "REFERENCE POINTS" (Fields 12-15 and 34-37). When a unit is in transit behind or in reserve out of the front line, a value of 0.0 is provided under "Width of Front." The front and rear of a unit march column may also be provided under "Left Flank" and "Right Flank." When information in sources is available, the route of a march column is identified by grid coordinates recorded under "REFERENCE POINTS." Data in "REFERENCE POINTS" is also used to describe the outline of a unit's "goose egg" position when it is in reserve. In cases where information in sources on a unit's location in reserve was minimal, only one or two "REFERENCE POINTS" may be used to identify the unit's location.

"SEPARATE ELEMENTS" (Fields 30-33): Data entered under "SEPARATE ELEMENTS" (using grid coordinates) describes the location of major elements of a unit not in the immediate vicinity of the main body of the unit, such as detached organic elements of the unit,or units in transit to the main body of the unit. When necessary, amplification of data recorded under "SEPARATE ELEMENTS" is provided under "Remarks."

"Total Time In Combat," "Total Time Not In Combat," and ancillary fields (Fields 16-21): Data entered in Fields 16-21 records the duration and time periods of a unit's participation in combat or it's time out of combat. Data is provided in these fields when available in sources.

"Major Decisions/Redirections," "Other Significant Events," and "Remarks" (Fields 22, 28, and 29): These are text fields provided for explanation of a unit's activities and any interpretive remarks to clarify data provided in other fields. Information is provided in these fields when it is available or pertinent to the understanding of data entered in any other field. In practice, due to limitations of text space in the computer data format, the use of the fields is interchangeable, i.e., they are all employed to record any textual information on a unit's activities, location, mission, displacement, etc.

"Unit Commander Data" (Field 23): This is a text field. Entered under "Unit Commander Data" are the name and rank of the commander of a unit. When a change in command occurs, it is also recorded under "Unit Commander Data." German commanders' rank titles are used for commanders of German units.

"Fortifications" (Field 24): This is a text field. Recorded under "Fortifications" is information on permanent and field fortifications maintained and occupied by a unit or other units of the same nationality, which are located in the vicinity of the unit. Information is provided in this field when available in sources.

"Obstacles" (Field 25): This is a text field. Recorded under "Obstacles" is information on permanent and field fortifications of a unit's opponents, or natural obstructions such as woods, rivers, and steep terrain in the vicinity of the unit. Information is provided in this field when available in sources.

"Surface Trafficability" (Field 26): This is a text field. Recorded under "Surface Trafficability" is information describing the condition of the ground and roads in the vicinity of the unit. Information is provided in this field when available in sources.

"Weather" (Field 27): This is a text field. Recorded under "Weather" is information on climatic conditions for the 24 hour period on the indicated date and in the vicinity of the unit. Information is provided in this field when available in sources. It is frequently taken verbatim from sources and is thus recorded in the format or style of the source.

"Sources" (No field number): Each unit has a unique corresponding Reference Data Base record which lists all sources used to compile unit location data for that unit. The Reference Data Base record is identified by a unique alphanumeric code, which is also entered under "Sources" in the Unit Location Data Base records of the corresponding unit. Attachment 1 contains a list with the names of units and the alphanumeric codes of their corresponding Reference Data Base records. These are the same codes entered under "Sources" in the Unit Location Data Base records. Note that the specific sources used for each item of information are not provided in the relevant Reference Data Base records. Instead, sources and the page numbers consulted in them are listed as the reference material for "All" data. This format proved necessary because of the impracticability of referencing every piece of data in the Unit Location Data Base, a task which would have resulted in the creation of an excessively large series of reference records for the Unit Location Data Base.

The ACSDB Air Data Base

INTRODUCTION

The Ardennes Campaign Simulation Data Base (ACSDB) Air Data Base contains information on tactical air sorties conducted by United States Army Air Forces (USAAF), Royal Air Force (RAF), and German Luftwaffe air units over the Ardennes battle area. This paper explains the general approach used to generate the ACSDB tactical air data. It is divided into four major sections. The first defines the data fields used in the Air Data Base, the second addresses the generation of US air data, the third describes British air data, and the fourth describes German air data. Data recorded in the Air Data Base required both the least and most amount of data estimation of any of the ACSDB data bases. Data on Allied (US and British) air sorties was taken directly from daily operations reports, while Luftwaffe data, for which primary source records are nearly non-existent, was almost completely estimated.

Full bibliographic annotation of all sources referenced in this paper are contained in the Bibliography Data Base.

"Day (MM/DD/YY)" (Field 1): Provided under Field 1 is the date in month-day-year format of the tactical air sorties on which information is recorded.

"# of Sorties flown" (Field 2): Provided under Field 2 is the number of sorties flown. A sortie is defined as one aircraft flying one mission.

"Aircraft Type" (Field 3): The generic type of aircraft flying all sorties counted in Field 2 is recorded in Field 3. Aircraft types are identified as:

- B (Bomber)
- F (Fighter)
- FB (Fighter-bomber)
- NF (Nightfighter)
- T (Transport)

"Aircraft Name" (Field 4): The specific aircraft model flying all sorties counted in Field 2 is identified in Field 4. Aircraft variants or sub-models are not identified, as data sources generally do not identify the variants and the differences in them are fundamentally insignificant. British aircraft are identified by name; US and German aircraft are identified by the alphanumeric designation for the aircraft. Aircraft models recorded in the Air Data Base are as follow:

- A-26 (Bomber)
- A-20 "
- Ar-234 "
- B-17 "
- B-24 "
- B-25 "
- B-26 "
- Boston "
- Halifax "
- Ju-88 "
- Lancaster "
- Mitchell "

- Me-109 (Fighter)
- P-38 "
- P-51 "
- Spitfire "

- Fw-190 (Fighter-bomber)
- Fw-190D "
- Me-262 "

- Mosquito "
- P-47 "
- Tempest "
- Typhoon "

- P-61 (Nightfighter)

- C-47 (Transport)
- Ju-52 "

"General Location of Mission" (Field 5): The geographical location of all sorties counted in Field 2 is recorded under Field 5. This is most often the area described or identified in the record, such as a town, road junction, etc.

"Airbase of Origin" (Field 6): Recorded under Field 6 is the name of the airbase of origin of all sorties counted in Field 2. Attachment 1 of this paper contains information on locations of US airbases from the Final Report of the Chief Engineer, European Theater of Operations. Locations of British and German airbases are also provided in a later section of this paper.

"Specific Location of Mission" (Field 7): Provided under Field 7 is the location of all sorties counted in Field 2, recorded in the Universal Transverse Mercator (UTM) four- or six-digit grid coordinates used on the 1943 Geographical Section, General Staff maps produced by the Army Map Service. These maps are:

Sheet #	Name
8	Brussels
9	Liege
12	Namur
13	Marche
16	Mezieres
14	Arlon
R-1	Koeln
S-1	Bonn
T-1	Trier
V-1	Neunkirchen

For purposes of the ACSDB, boundaries were established to define the Ardennes area within which all tactical air sorties that occurred were to be recorded in the Air Data Base. These boundaries are:

- On the east, the Rhine and Moselle Rivers.

- On the north, an east-west line running from Mulheim (F5064) to Mechelen (K5864).

- On the west, the Meuse River.

- On the south, a west-east line running from Flize (O8626) to Wasserbillig (Q1125).

In the initial stages of the Ardennes Campaign, the US and British air forces carried out significant air operations of a strategic nature within 40 kilometers of the 16 December forward edge of the battle area (FEBA), including attacks on Cologne, Dusseldorf, Bonn, Trier, and other cities. These attacks included raids on industrial targets and the railroad transportation network. As the Ardennes Campaign continued, Allied air operations against the German rail system intensified and struck targets within 15 kilometers of the 16 December FEBA. Attacks on the rail system also extended east of the Rhine and Moselle Rivers well into the heart of Germany. Due to the large numbers of Allied air operations against industrial and rail targets, it was decided to exclude these kinds of air missions, which were primarily of a strategic nature, and focus the air data research effort on tactical and other types of mission. The effects of the strategic Allied air effort is represented by the arrival, or failure to arrive, of German ground units and supplies. However, note that the Air Data Base does include interdiction missions against German ground units and supplies arriving by rail, once they have been detrained. Specific criteria for inclusion of Allied air operations in the ACSDB are as follow:

- They are attacks on detrained units.
- They are of an unspecified purpose.
- They are attacks on non-rail transported logistics.
- They are attacks on any stationary target such as a depot or positions that directly support combat operations.
- They are attacks on detrained units at rail heads.

Allied air operations are excluded if:

- They are against entrained units.
- They are against industrial targets.
- They are fighter escort missions for missions that would not otherwise be included.
- They are exclusively attacking the railroad facilities of a rail head.
- They are attacking a railroad bridge, a railroad tunnel, or other components of the railroad infrastructure.
- Aircraft recalled due to weather are excluded. Aircraft turning back due to weather are excluded from the mission total even if the mission was completed by other aircraft in that flight. Information is usually provided under "Remarks" (Field 11).

All photoreconnaissance, reconnaissance, or other missions without an identifiable location are excluded. A photoreconnaissance, reconnaissance, or other type of mission

against an identifiable target in the battle area is included. Aircraft that spend only part of the time in the Ardennes Campaign area are recorded, and their activity outside of the area is noted under Field 11. Aircraft that have mechanical problems or failures are included, and are usually described under "Remarks."

Criteria for inclusion of German Luftwaffe air operations in the Air Data Base are not as exact as those for Allied air operations because of the nature of the available German records and the methodology used to estimate daily German air data. Essentially, any German air operation over the Ardennes Campaign area was included in the ACSDB.

"Aircraft Nationality" (Field 8): Recorded under Field 8 is the nationality of all sorties counted in Field 2. "A" = US (American), "B" = British, and "G" = German.

"Mission" (Field 9): Recorded under Field 9 is a brief description of the type of mission conducted by all aircraft making the sorties counted in Field 2. The descriptions were derived from those used in the historical records. These include:

- Aerial Resupply: Supply of ground units by aircraft.
- Alert Flight: Fighters airborne to intercept enemy aircraft.
- Attack: Ground attack mission of unidentified nature (bombs, rocket, strafing, etc.).
- A/R: Armed reconnaissance.
- A/R-Night: Armed reconnaissance at night.
- Blind Bombing - Night: Bombing mission conducted without visual target identification/acquisition at night.
- Blind Bombing - Radar: Bombing mission conducted with radar target acquisition.
- Bombing: Includes bombing missions using high-explosive bombs, fragmentation bombs, napalm, and rockets.
- Escort: Fighter escort of a bombing mission.
- Fighter Sweep: Fighter patrol of an area.
- Immediate Support: British term for "Support" (q.v.).
- Intercept: Fighters directed to intercept enemy aircraft.
- Intruder - Night: Night intercept/fighter sweep mission.
- Pathfinding: Mission flown by an aircraft to mark bombardment target area with flares.
- Paradrop: Supply of ground units by parachute.
- Patrol: Fighter patrol of an area.
- Patrol - Night: Fighter patrol of an area at night.
- Photo Recon: Photo reconnaissance.
- Scramble: Fighters taking off to defend airfield.
- Support: Tactical support of ground units. Unit is identified when known.
- Weather Recon: Weather reconnaissance mission.

"Sources" (Field 10): Recorded under Field 10 is an alphanumeric code which matches an entry under "Source Number" in the Reference Data Base record which lists the sources used for the corresponding Air Data Base record. All German air records are referenced using a code beginning with "AG" followed by three arabic numerals ("AA001," "AA012," etc.) All British air records are referenced using a code beginning with "AB" followed by three arabic numerals ("AB001," "AB015," etc.).

Entries under Field 10 for US air records use a coding systems which corresponds to the unit of the aircraft conducting all sorties recorded in Field 2.

- AA001 series (AA001-AA100): XXIX Tactical Air Command aircraft.
- AA200 series: XIX Tactical Air Command aircraft.
- AA300 series: IX Tactical Air Command aircraft.
- AA400 series: 9th Bombardment Division.
- AA700 series: Eighth Air Force aircraft.
- AA900 series: 9th Troop Carrier Command aircraft.

"Remarks" (Field 11): Additional information which amplifies data in any other field is recorded in text format under Field 11.

Abbreviations used in Field 11 and other fields of the Air Data Base, as derived from historical sources, include:

- A/C: Aircraft
- AI: Air intercept (visual or radar).
- E/A: Enemy aircraft.
- Gp: Group.
- MET: Motorized enemy transport.
- M/T: Motor transport.
- NFS: Night fighter squadron.
- NRO: No results observed.
- NYR: Not yet returned.
- PFF: Pathfinder force.
- R/V: Rendezvous.
- Spare a/c: An aircraft that flies along with a formation of aircraft on a combat mission, and which replaces any aircraft that aborts before reaching enemy territory
- Sq: Squadron.
- T/G: Target.
- U/I: Unidentified.
- (Unk): Unknown -- refers to lost airplane whose status is not known, i.e., destroyed or not yet returned, or refers to unknown cause of loss.
- Wg: Wing.
- Window a/c: Aircraft dropping chaff (foil strips) to interfere with radar